Comparison of DoA Algorithms for MIMO OFDM Radar

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***Abstract* — This paper presents results for the determination of the direction of arrival (DoA) problem for a MIMO OFDM radar system using some of the most popular techniques: Multiple Signal Classification (MUSIC), Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT), Minimum Norm (Min-Norm), Minimum Variance Distortionless Response (MVDR). The performance of the algorithms is evaluated using two different metrics, namely the performance achievable in terms of resolution and the probability of target distinction, since, for example, in automotive systems, not only the resolution but also the correct distinction of the number of targets can be crucial. The results presented in the paper show that the Min-Norm algorithm is a better option when used for applications that require reliable distinction of targets.**

***Keywords* — MIMO, OFDM, RADAR, RadCom.**

I. Introduction

Radar detection and wireless communications are among the most prominent radio applications. However, they were studied and developed independently in most scenarios. A system that uses dual devices which can perform both concurrent reflectometry and common radar functions with communication, has significant advantages in terms of cost, size, performance and spectrum used [1].These common radar/communication systems (RadCom) use the same signals and hardware to perform the detection and communication simultaneously. These systems have the potential to be employed for area surveillance, search and rescue, and intelligent transportation.

The use of OFDM for radar was first proposed in [2]. Preliminary studies of the integration of radar and communication functionalities were carried out in [3] and several additional contributions followed. For example in [4] the authors analyse the performance of a four-station network with radar and communications and evaluate the throughput rate. In [5] the authors report a study for an imaging radar using the same radio resources for transmitting the acquiring images. A major step in the applicability of the RadCom system was presented in [6], where a more efficient and simple approach to OFDM radar processing was proposed. This processing technique operates directly on the modulated symbols instead of the baseband signal, overcoming the disadvantages of classical correlation-based processing that had been proposed so far.

One important issue common to both radar and communication systems, employing multiple antennas, is the determination of the direction-of-arrival (DoA) of the received waveforms. While in traditional radar mechanical scanning is frequently used, for future applications, namely for the intelligent transportation system, where the angular estimate of conventional radars is not feasible due to the space limitations and also due to the velocity of the vehicle in which the radar is located. The use of Multiple Input Multiple Output (MIMO) for DoA estimation provides a superior performance in relation to the conventional phase array, having better resolution in DoA estimation [7].

Studies of DoA estimation techniques for OFDM radars have already been performed in [7], [8], [9]. In [8] the performance of the MUSIC technique for separation of close targets is evaluated. However, no comparative study among possible DoA estimations techniques was performed. This paper compares the performance of MUSIC, ESPRIT, Min-Norm and MVDR [10] for DoA estimation in RadCom systems, with a focus on the evaluation of their performance for target separation. As will be shown, the MUSIC algorithm does not achieve a satisfactory performance for separation of close targets, failing to distinguish. For vehicle radar purposes this type of problem compromises the reliability of the system.

II. System Model

The time domain OFDM transmitted signal composed by N subcarriers and M OFDM symbols is expressed by:

$$x(t) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \mathbf{S}(m,n) e^{2\pi j n \Delta f t} \text{rect}\left(\frac{t - mT}{T}\right) \quad (1)$$

where Δf denotes the subcarrier spacing and $\mathbf{S}(m,n)$ the transmit data, modulated with a digital complex-modulation technique. In this paper we will consider phase-shift keying (4-PSK). In the presence of K reflective targets at a distance R_k of the RadCom system and relative velocities of v_k, which results in the Doppler frequency f_{Dk}, the received OFDM symbol becomes [8]:

$$y(t) = \sum_{k=1}^{K} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \mathbf{S}(m,n) e^{j2\pi f_{Dk} t} e^{j2\pi n \Delta f (t - \frac{2R_k}{c})} + \eta(t) \quad (2)$$

and therefore, the received modulation symbols $\hat{\mathbf{S}}(m,n)$ are

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$$\hat{\mathbf{S}}(m,n) = \sum_{k=1}^{K} \mathbf{S}(m,n) e^{j2\pi f_{Dk}} e^{-j2\pi n \Delta f \frac{2R_k}{c}} + \eta(m,n) \quad (3)$$

where η is additive complex white Gaussian noise (AWGN).

Consider a MIMO radar system with M_T transmit antennas and M_R receive antennas, illuminating K uncorrelated targets from the directions θ_k, $k = 1, 2, ..., K$ and employs the OFDM signal. The transmitter and receiver of the MIMO radar are assumed to be the same, with uniform inter-element spacing d of a half wavelength ($\lambda/2$). Since $\mathbf{S}(m,n) = [\mathbf{S}_1(n,m), ..., \mathbf{S}_{M_T}(n,m)]$ is the set of symbols transmitted by the M_T transmit antennas, the received set of complex modulation symbols $\hat{\mathbf{S}}(m,n) = [\hat{\mathbf{S}}_1(n,m), ..., \hat{\mathbf{S}}_{M_R}(n,m)]$ by the M_R receive antennas can be described by the equation:

$$\hat{\mathbf{S}}(m,n) = \sum_{k=1}^{K} [\mathbf{a}_T(\theta_k) \otimes \mathbf{a}_R(\theta_k)] \sum_{p=1}^{M_T} \mathbf{S}_p(m,n) \quad (4)$$

$$e^{j2\pi f_{Dk}} e^{-j2\pi n \Delta f \frac{2R_k}{c}} + \eta_q(m,n)$$

$$= \sum_{k=1}^{K} \mathbf{A}(\theta_k) \sum_{p=1}^{M_T} \mathbf{S}_p(m,n) e^{j2\pi f_{Dk}} e^{-j2\pi n \Delta f \frac{2R_k}{c}} + \eta_q(m,n) \quad (5)$$

with $\mathbf{A}(\theta_k) = [\mathbf{a}_T(\theta_k) \otimes \mathbf{a}_R(\theta_k)]$, where $\otimes$ represents the Kronecker product and $\mathbf{a}_T(\theta_k)$ and $\mathbf{a}_R(\theta_k)$ are the steering vectors of the transmitter and receiver for the kth target, given by:

$$a_T(\theta_k) = \left[e^{j2\pi d 1 \sin\theta_k/\lambda}, \cdots, e^{j2\pi d M_T \sin\theta_k/\lambda} \right]^T \quad (6)$$

$$a_R(\theta_k) = \left[e^{j2\pi d 1 \sin\theta_k/\lambda}, \cdots, e^{j2\pi d M_R \sin\theta_k/\lambda} \right]^T \quad (7)$$

III. Radar Processing

The estimate of the range and Doppler shift is performed through a spectral estimate; comparing the transmitted signal $\mathbf{S}(m,n)$ with the received signal $\hat{\mathbf{S}}(m,n)$ generates the frequency domain channel transfer function D:

$$\mathbf{D}_h(m,n) = \frac{\hat{\mathbf{S}}(m,n)}{\mathbf{S}(m,n)}, \qquad h = 1, 2..., M_T \times M_R \quad (8)$$

$$= \sum_{k=1}^{K} \mathbf{A}(\theta_k) e^{j2\pi f_{Dk}} e^{-j2\pi n \Delta f \frac{2R_k}{c}} + \eta(m,n) \quad (9)$$

The range and velocity parameters can be obtained from a 2D-DFT [6], [8]. The impulse response of the channel containing the range profile of the object can then be determined by taking an IDFT:

$$\mathbf{Z}_h(i) = \mathrm{IDFT}[\mathbf{D}_h(n)], \quad i = 0, 1..., N-1 \quad (10)$$

The Doppler shift can be estimated taking the FFT through the time axis of the corresponding channel transfer function:

$$\mathbf{Z}_h(j) = \mathrm{DFT}[\mathbf{D}_h(m)], \quad j = 0, 1..., M-1 \quad (11)$$

IV. DoA Estimation

For the processing of the DoA estimation techniques used in this paper, the signal processing can be applied directly to received signals. Then for this case the velocity profile of the channel information matrix $\mathbf{D}(m) = [\mathbf{D}_1(m), , ..., \mathbf{D}_{M_T \times M_R}(m)]$ was used to calculate the covariance matrix $\mathbf{R} = \frac{1}{M} \sum_{m=0}^{M-1} \{\mathbf{D}(m)\mathbf{D}(m)^T\}$. This covariance matrix $\mathbf{R}$ can be written as:

$$\mathbf{R} = \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^H + \mathbf{U}_n \mathbf{\Lambda}_n \mathbf{U}_n^H \quad (12)$$

where $\mathbf{U}_s$ represents the signal subspace consisting of the largest K eigenvalues of $\mathbf{R}$ and $\mathbf{U}_n$ represents the noise subspace with the remaining $M_R \times M_R - K$ eigenvectors. $\mathbf{\Lambda}_s$ and $\mathbf{\Lambda}_n$ are diagonal matrices containing corresponding eigenvalues. The DoA are defined by maximum values in the spectrum using these subspaces and the correlation matrix, as defined below.

A. MVDR

MVDR uses the signal correlation matrix for DoA estimation. The technique attempts to minimize the power contributed by noise and unwanted interference, while maintaining a fixed gain in the look direction, and the nulls in other directions to reject other signals [10]. The peaks in the MVDR spectrum occur whenever the direction vector is orthogonal to the noise subspace. The angular spectrum MVDR is defined by

$$P_{MVDR} = \frac{1}{\mathbf{A}^H(\theta)\mathbf{R}^{-1}\mathbf{A}(\theta)} \quad (13)$$

B. MUSIC

MUSIC is one of the most recent methods proposed and a very popular method for estimation of arrival direction based on subspace techniques. The spectrum of MUSIC is given by:

$$P_{MUSIC} = \frac{1}{\mathbf{A}^H(\theta)\mathbf{U}_n\mathbf{U}_n^H\mathbf{A}(\theta)} \quad (14)$$

MUSIC estimates all possible direction vectors $\mathbf{A}(\theta)$ to find those that are perpendicular to the space covered by the noise eigenvectors, since the direction vectors corresponding to the DoA are in the subspace of the signal and are orthogonal to the noise subspace [10].

C. Min-Norm

The Min-Norm algorithm can improve the performace of the MUSIC algorithm. The estimation function uses the new noise subspace which is a linear combination of the noise subspace $\mathbf{U}_n$, therefore, the new noise subspace is orthogonal to the signal subspace. Its spectrum is given by [10]:

$$P_{MN} = \frac{1}{\mathbf{A}^H(\theta)\mathbf{U}_n\mathbf{U}_n^H\mathbf{W}\mathbf{U}_n\mathbf{U}_n^H\mathbf{A}(\theta)} \quad (15)$$

with the vector $\mathbf{W} = ww^T$ where w is equal to the first column of an identity matrix $M_T M_R \times M_T M_R$.

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D. ESPRIT

ESPRIT is another subspace-based DoA estimation algorithm, but it does not perform a search on all possible direction vectors to estimate DoA, which reduces processing and memory usage. ESPRIT is based on the rotational invariance of the subspace of the incident signals expanded by two responses shifted from each other by the $\mathbf{\Phi}$ matrix.The system response is then composed of two subsets whose values are related to each other by the displacement vector, so there must be a unique nonsingular matrix $\mathbf{T}$ such that $\mathbf{U}_s$ can be decomposed into two subspaces $\mathbf{U}_1$ and $\mathbf{U}_2$ [10]:

$$\mathbf{U}_s = \begin{bmatrix} \mathbf{U}_1 \\ \mathbf{U}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{AT} \\ \mathbf{A\Phi T} \end{bmatrix} \tag{16}$$

and

$$\mathbf{U}_1\mathbf{\Psi} = \mathbf{U}_2 \Longrightarrow \mathbf{AT\Psi} = \mathbf{A\Phi} \tag{17}$$

$\mathbf{\Phi}$ and $\mathbf{\Psi}$ are related via an eigenvalue-preserving similarity transformation.

$$\mathbf{\Psi} = \mathbf{T}^{-1}\mathbf{\Phi T} \tag{18}$$

The diagonal elements (or eigenvalues) of $\mathbf{\Phi}$ are equal to the eigenvalues of the transformation matrix $\mathbf{\Psi}$ that relates $\mathbf{U}_1$ and $\mathbf{U}_2$. then the arrival angles can be estimated by the eigenvalues (λ_Φ) of $\mathbf{\Phi}$:

$$\theta_k = \sin^{-1}\left(\frac{\lambda}{2\pi d}\arg(\lambda_{\Phi k})\right) \tag{19}$$

where $\arg(\xi)$ is the function that returns the argument of the complex number ξ.

V. Simulation and Analysis of Results

Numerical simulations were performed to compare the performance of DoA techniques: MVDR, MUSIC, Min-Norm and ESPRIT. The estimation capacity of these techniques was evaluated for a single target and for two targets (target A and target B) with a difference of 3 degrees in the presence of white Gaussian noise. Note that in this paper, we suppose that K is already known. The performances are measured by the Root Mean Square Error (RMSE).

The simulated radar system has a carrier frequency of 24 GHz, bandwidth of 93.1 MHz, with 1024 subcarriers spaced by 90.909 kHz. The simulation evaluates 256 4-PSK symbols with a duration per symbol of 12.375 μs. The system has 3 transmitting antennas and 3 receiving antennas spaced by $d = \lambda/2$. Details of the system parameterization are shown in [6]. In the simulations, the angles are random. The velocity and position of targets A and B are V_A = 7 m/s V_B = 6 m/s, R_A = 10 m and R_B = 10 m respectively.

Fig. 1 shows the DoA RMSE performance of the algorithms with respect to the signal-to-noise ratio for the scenario with one target.The Monte Carlo estimations were performed through 8000 simulations. With the results presented it is possible to verify that the MUSIC technique has the best performance, together with the MVDR technique. The Min-Norm technique and ESPRIT are the ones that present the worst resolution. The ESPRIT technique is the worst for values above 10 dB.

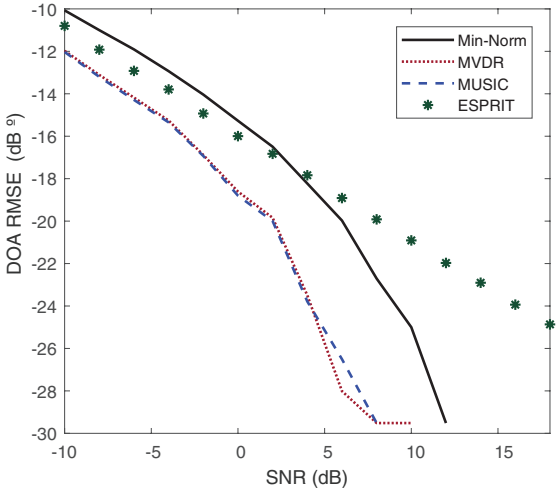


Fig. 1. RMSE for the resolution of one target versus SNR.

In the simulation shown in Fig. 2 we present the results for DoA RMSE for the scenario with two targets with a difference of 3°. The angles of the two targets were randomly varied from $-75°$ to 75°, always remaining 3 degrees between the two targets. In the case where the angle was not determined we consider as a straight line the direction of the unknown target (0°). The figure clearly shows that the performance of the MUSIC and MVDR algorithm is compromised, with the Min-Norm and ESPRIT algorithms now having the best performance.

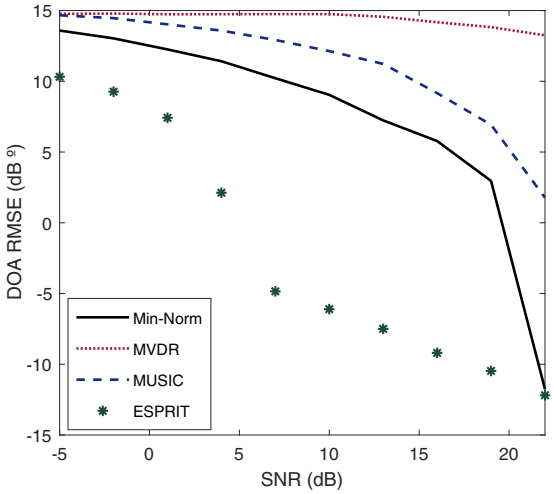


Fig. 2. RMSE for the resolution of two target versus SNR.

To evaluate the ability to distinguish targets by the algorithms, the Fig. 3 is presented, where the probability of correct distinction of the number of sources is presented. This figure shows the results of the MVDR, MUSIC and

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Min-norm techniques; since the ESPRIT technique always provides the number of angle estimates equal to the number of targets reported, different from the other three methods which may show divergence, its results were not taken into account here. The results of this simulation show that the Min-Norm technique has a much higher resolution capability than other methods, for example having a target discrimination probability of 60% at an SNR of -5 dB against 10% of MUSIC and 0% of MVDR.

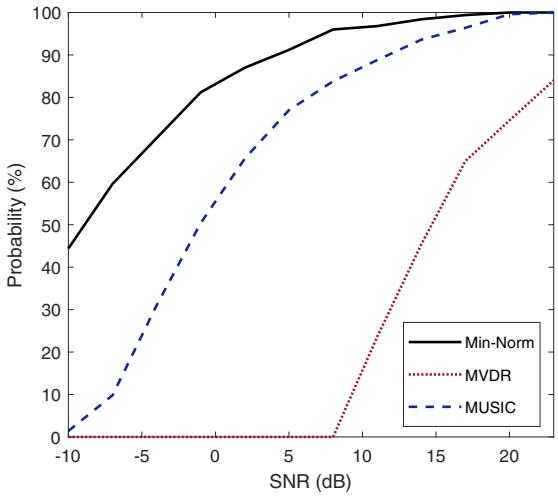


Fig. 3. Distinction probability versus SNR.

The Figs. 4 shows the comparison of the results of DoA spectrum with two closely located targets. In this simulation, the SNR is 0 dB and the angles are 51 and 54 degrees. It is possible to see that only the Min-Norm technique was able to distinguish the two targets, while MUSIC and MVDR presented only one DoA estimation peak.

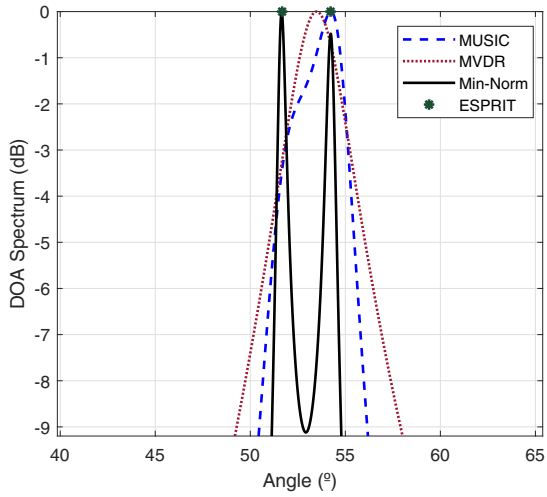


Fig. 4. DoA spectrum with two closely located targets

VI. Conclusion

This paper presents the results of the direction-of-arrival estimation based on a MIMO OFDM radar system using some of the most popular techniques for estimation of DoA: MVDR, MUSIC, Min-Norm and ESPRIT. The results of the simulation showed that the method that presents the best resolution in determining a target is the MUSIC algorithm, followed by Min-norm and ESPRIT, and the one with the worst resolution is the MVDR. However, in the simulations performed with two separate sources at a small angular distance, we showed that the Min-Norm technique presents a much better estimation probability than the other techniques, having a much greater capacity to distinguish the two targets, followed by ESPRIT. The MUSIC and MVDR techniques do not present a good performance for the distinction of the two targets, presenting a much lower probability of estimation when compared to Min-Norm, with exception only for situations with high SNRs. Since in some applications, such as automotive systems, the distinction of the targets may be more important than the decimal resolution, the Min-Norm technique may be a better option to use.

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